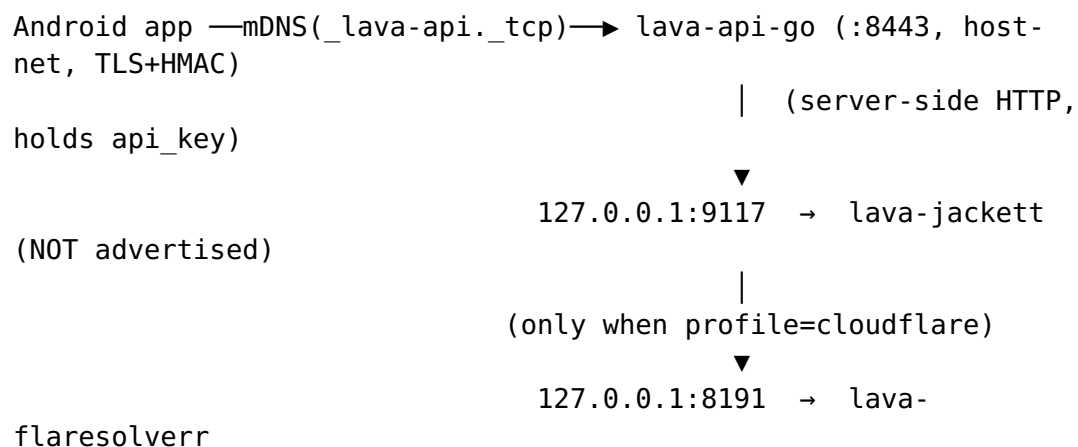


Jackett Torznab Sidecar — Deployment & Validation Guide

Status: deployment-ready (compose fragment + .env.example keys + validation script authored; no container booted in this authoring pass). Companion: [jackett-local-stack-research.md](#) (decision dossier) and [jackett-local-stack-implementation.md](#) (Go Torznab client). Operator decision implemented: **Jackett as a Torznab sidecar in the Lava local stack, fronted by lava-api-go, mDNS-discovered, NOT bundled in the APK.**

This guide covers how to start the sidecar, validate it end-to-end with one command, perform the one-time (stateful, partly manual) Jackett indexer setup for IPTorrents (with FlareSolverr), and how the Android app reaches it.

1. Topology



- The app discovers **exactly one** service type — `_lava-api._tcp` (= `lava-api-go`, default `LAVA_API_MDNS_TYPE` in `lava-api-go/internal/discovery/mdns.go`). Jackett has no mDNS advertisement and is **never** put on the LAN.
- `lava-api-go` runs `network_mode: host` (mDNS requires host net). A host-net container **cannot** resolve the bridge DNS name `lava-jackett`, so the sidecar's port is published to the **host loopback** (`127.0.0.1:9117`), and `lava-api-go` reaches it at

http://127.0.0.1:9117. Loopback bind keeps it off the LAN (same posture as lava-postgres's 127.0.0.1:8432).

- **FlareSolvrr** (headless Chromium) is only needed for Cloudflare-protected trackers (notably **IPTorrents**). It is heavy, so it lives behind the cloudflare profile and is OFF by default. Jackett reaches it over the lava-net bridge at http://lava-flaresolvrr:8191 (both are bridge members); the loopback publish of 8191 is only for the validation curl.

2. Files

File	Role
tools/lava-containers/docker-compose.jackett.yml	compose fragment — lava-jackett (+ optional lava-flaresolvrr) on lava-net, loopback-published
scripts/validate-jackett-sidecar.sh	one-command end-to-end readiness check (bring up → wait healthy → probe caps + flaresolvrr → teardown)
.env.example	placeholders for all LAVA_API_JACKETT_* + LAVA_JACKETT_* + LAVA_FLARESOLVERR_* keys
lava-api-go/internal/jackett/	the Go Torznab client lava-api-go uses (see implementation doc)
lava-api-go/internal/config/config.go	reads LAVA_API_JACKETT_* env (the consumer side)

3. Environment variables (§6.R — nothing hardcoded)

Copy .env.example → .env and fill these (the .env is gitignored):

Variable	Consumed by	Purpose / default
LAVA_API_JACKETT_ENABLED	lava-api-go	turn the Jackett route on (false)
LAVA_API_JACKETT_URL	lava-api-go	http://127.0.0.1:9117 (loopback — NOT a bridge name; see §1)
LAVA_API_JACKETT_APIKEY		

Variable	Consumed by	Purpose / default
	lava-api-go + healthcheck + validation script	the Jackett api_key (no default — generated on first run)
LAVA_API_JACKETT_DEFAULT_INDEXER	lava-api-go + healthcheck	indexer id for the caps probe (all)
LAVA_JACKETT_IMAGE	compose	Jackett image — pin by digest in real .env
LAVA_FLARESOLVERR_IMAGE	compose	FlareSolvrr image — pin by digest
LAVA_JACKETT_BIND_HOST / LAVA_JACKETT_HOST_PORT	compose	loopback bind (127.0.0.1 / 9117)
LAVA_FLARESOLVERR_BIND_HOST / LAVA_FLARESOLVERR_HOST_PORT	compose	loopback bind (127.0.0.1 / 8191)
LAVA_JACKETT_CONFIG_DIR	compose	gitignored / config host volume (./.jackett-config) — \$6.H secrets store
LAVA_JACKETT_PUID / _PGID / _TZ	compose	LinuxServer user/group/ timezone
LAVA_FLARESOLVERR_LOG_LEVEL	compose	FlareSolvrr verbosity (info)

The api_key is the SAME value for the Go service AND the container healthcheck AND the validation script — there is no parallel name to keep in sync. It is a \$6.H secret: server-side only, never committed (.gitignore excludes .jackett-config/), never sent to the device.

4. One-command end-to-end validation

After the `api_key` is set in `.env` (see §5), the main agent runs:

```
# Jackett only (RuTracker / RuTor / NNMClub indexers – no
Cloudflare):
scripts/validate-jackett-sidecar.sh

# Jackett + FlareSolverr (the IPTorrents / Cloudflare path):
scripts/validate-jackett-sidecar.sh --cloudflare
```

The script auto-detects podman (preferred) then docker, brings the sidecar up, waits for the compose healthchecks to go HEALTHY, then independently re-probes from the host (the §6.B/§6.J discipline — not trusting `State==running`):

- **Jackett:** GET `.../results/torznab/api?t=caps&apikey=...` → expects HTTP 200 + parseable Torznab XML, `api_key` not rejected. (Verbatim body printed.)
- **FlareSolverr** (`--cloudflare`): POST `/v1 {"cmd":"sessions.list"}` → expects HTTP 200 + `"status":"ok"`. (Verbatim body printed.)

Exit 0 = READY, 1 = a probe failed (logs tailed), 2 = config error (no runtime / missing `api_key`). The stack is torn down on exit unless `--keep`.

5. One-time Jackett setup (stateful — partly MANUAL)

Jackett's configuration lives in the gitignored `/config` volume and persists across restarts. These steps cannot be fully automated (the dossier's Open Question Q3 — admin HTTP vs per-`/config` JSON — is unresolved):

5.1 Generate the `api_key` (required before validation passes)

1. Start Jackett once with the config volume mounted (e.g. the validation script with `--keep`, or `docker compose -f tools/lava-containers/docker-compose.jackett.yml --profile jackett up -d` after the network exists). On first run with an empty `/config`, Jackett writes a fresh `api_key` into `${LAVA_JACKETT_CONFIG_DIR}/Jackett/ServerConfig.json` under the `APIKey` field (or read it from the WebUI top-right "API Key" field at `http://127.0.0.1:9117`).
2. Copy that value into `.env` as `LAVA_API_JACKETT_APIKEY=...`.
3. Re-run the validation script — the caps probe now authenticates. On a zero-indexer config the caps body is an

empty <caps/>; the script accepts this as "surface live + api_key valid". A real search needs §5.2.

5.2 Add the IPTorrents indexer (with FlareSolvrr) — MANUAL WebUI step

IPTorrents sits behind Cloudflare, so FlareSolvrr MUST be running and Jackett MUST be pointed at it:

1. Bring up the cloudflare profile so lava-flaresolvrr is running.
2. In the Jackett WebUI (<http://127.0.0.1:9117>) → top-right **gear** / **Settings**: set **FlareSolvrr API URL** to <http://lava-flaresolvrr:8191> (bridge DNS name — Jackett reaches FlareSolvrr over lava-net). Save.
3. **Add indexer** → search "IPTorrents" → **+** → enter the IPTorrents credentials (these are a §6.H secret; they live ONLY in the gitignored /config volume, never in tracked source). Save.
4. (Optional) Add RuTracker / RuTor / NNMClub indexers the same way — they do not need FlareSolvrr, so the jackett-only profile suffices for them.

Per §6.A: if the indexer-add is ever automated via Jackett's admin HTTP API, that automation MUST gain a contract test asserting the request/flag shape (mirroring lava-api-go/[tests/contract/healthcheck_contract_test.go](#)) and pin the Jackett version, because the admin API surface is version-sensitive.

6. How the Android app discovers + uses it

The app never sees Jackett. The flow:

1. App auto-discovers lava-api-go via mDNS `_lava-api._tcp` (existing discovery; unchanged by this work).
2. App calls lava-api-go's Torznab-backed search route over TLS+HMAC.
3. lava-api-go (holding the api_key server-side) calls Jackett at <http://127.0.0.1:9117/api/v2.0/indexers/<id>/results/torznab/api>, parses the Torznab XML via internal/jackett, and returns clean results to the app.
4. Downloads: Jackett answers /dl/ links with HTTP 302 → magnet:; the Go client captures the Location verbatim (does NOT auto-follow) and returns a real magnet or .torrent bytes (see the implementation doc).

Mapping to IPTorrentsConfig

The IPTorrents provider plugin (core/tracker/iptorrents/.../IPTorrentsConfig.kt) resolves the **lava-api-go sidecar** base URL (NOT Jackett directly — the app talks only to lava-api-go), in priority order, from:

1. an explicit override,
2. the iptorrentsJackettBaseUrl JVM system property (set by the §6.G test task),
3. the IPTORRENTS_JACKETT_BASE_URL environment variable (the runtime value).

There is **no compile-time default** (§6.R): unconfigured → resolve() returns null → the feature fails honestly ("sidecar base URL not configured") rather than guessing a host. On Android the production value is injected via BuildConfig / runtime config in :core:tracker:client. So the chain is:

IPTORRENTS_JACKETT_BASE_URL → IPTorrentsConfig.resolve() → lava-api-go (Torznab route)

→ Jackett (LAVA_API_JACKETT_URL)

→ IPTorrents (via FlareSolverr)

7. What CANNOT be automated (honest manual list)

Step	Why it's manual	Mitigation
api_key into .env (§5.1)	Jackett mints the key on first run into its stateful /config; there is no pre-seeding	validation script refuses the placeholder and prints the exact remediation
IPTorrents indexer add + credentials (§5.2)	per-indexer config is WebUI/API state in /config; IPTorrents creds are §6.H secrets that must not enter tracked source	documented WebUI steps; future §6.A-contract-tested automation is owed (dossier Q3)
FlareSolverr URL setting in Jackett (§5.2)	a Jackett WebUI/config-file setting, not an env var Jackett reads	one-time WebUI step; persisted in /config
Image digest pinning	the real digest is environment-specific;	pin by digest in the real .env (§6.K) before any gate run

Step	Why it's manual	Mitigation
	.env.example carries the :latest placeholder	

Everything else — bring-up, healthcheck wait, caps + FlareSolverr probing, teardown — is fully automated by scripts/
validate-jackett-sidecar.sh.